Class 11: Protein Structure Prediction with AlphaFold

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Background

We saw last day that the main repository for biomolecular structure (the PDB database) only has ~250,000 entries.

UniProtKB (the main protein sequence database) has over 200 million entries!

The EBI AlphaFold Database

The EBI AlphaFold database contains lots of computed structural models. It is increasingly likely that the structure you are interested in is already in this database at < https://alphafold.ebi.ac.uk/ >

There are three major outputs from AlphaFold:

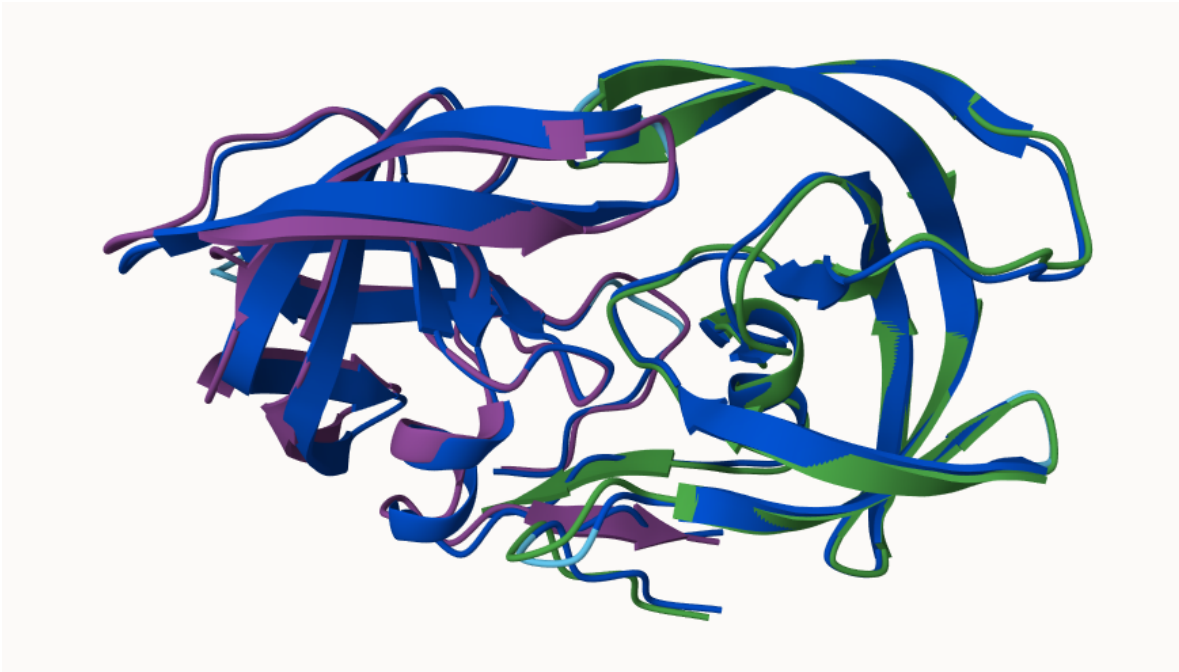
1. A model of structure in **PDB** format
2. A **pLDDT** score that tells you how confident the model is for a given residue in your protein (score above 70 is considered good)
3. A **PAE** score that tells you about protein packing quality

If you cannot find the matching entry for the sequence you are interested in AFDB or PDB then you can run AlphaFold yourself and generate your own structure predictions using ColabFold…

Generating Your Own Structure Predictions by Running AlphaFold

Use ColabFold at < https://github.com/sokrypton/ColabFold?tab=readme-ov-file >

Figure from AlphaFold:



Custom Analysis

We can read the AlphaFold results into R and do more quantitative analysis than just viewing the structure in Mol-star.

Read all the PDB models:

```
results_dir <- "hivpr_23119/"

pdb_files <- list.files(path=results_dir,
                        pattern=".pdb",
                        full.names = TRUE)

# Print out PDB file names
basename(pdb_files)
```

```
[1] "hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```r
library(bio3d)

# Read all data from Models and superpose/fit coords
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
hivpr_23119//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
.....

Extracting sequences

pdb/seq: 1   name: hivpr_23119//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4
pdb/seq: 2   name: hivpr_23119//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1
pdb/seq: 3   name: hivpr_23119//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5
pdb/seq: 4   name: hivpr_23119//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2
pdb/seq: 5   name: hivpr_23119//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3
```

```r
# library(bio3dview)
# view.pdbs(pdbs)
```

How similar or different are my models?

```r
rd <- rmsd(pdbs, fit=T)
```

```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

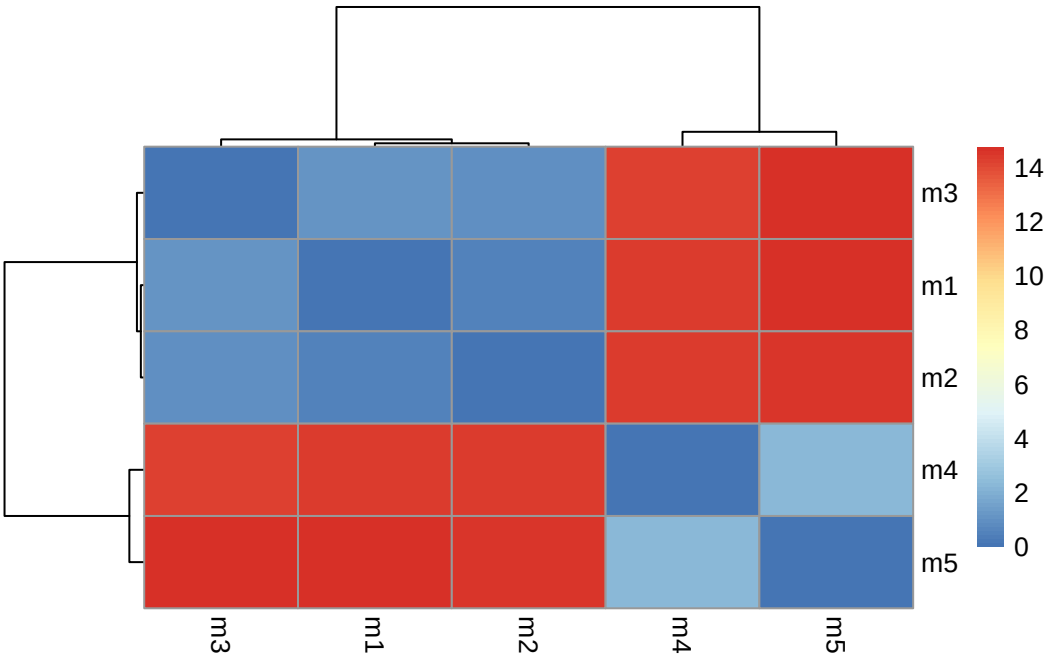
```r
range(rd)
```

```
[1]  0.000 14.754
```

Draw a heatmap of these RMSD matrix values:

```r
library(pheatmap)

colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```

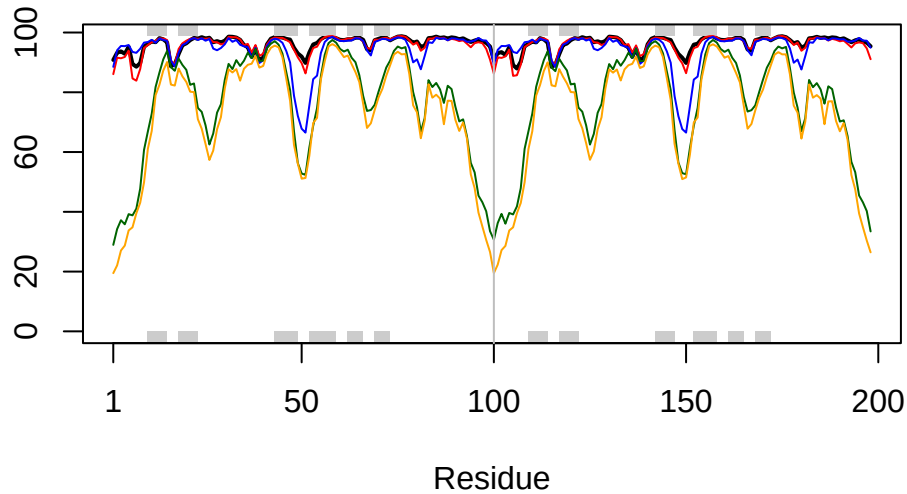


Plot the pLDDT values across all models:

```
# Reading a reference PDB structure
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```



We can improve the superposition/fitting of our models by finding the most consistent "rigid core" common across all the models. For this we will use the `core.find()` function:

```
core <- core.find(pdbs)
```

```
core size 197 of 198  vol = 9885.419
core size 196 of 198  vol = 6898.241
core size 195 of 198  vol = 1338.035
core size 194 of 198  vol = 1040.677
core size 193 of 198  vol = 951.865
core size 192 of 198  vol = 899.087
core size 191 of 198  vol = 834.733
core size 190 of 198  vol = 771.342
core size 189 of 198  vol = 733.069
core size 188 of 198  vol = 697.285
core size 187 of 198  vol = 659.748
core size 186 of 198  vol = 625.28
core size 185 of 198  vol = 589.548
core size 184 of 198  vol = 568.261
core size 183 of 198  vol = 545.022
core size 182 of 198  vol = 512.897
core size 181 of 198  vol = 490.731
core size 180 of 198  vol = 470.274
```

```
core size 179 of 198  vol = 450.738
core size 178 of 198  vol = 434.743
core size 177 of 198  vol = 420.345
core size 176 of 198  vol = 406.666
core size 175 of 198  vol = 393.341
core size 174 of 198  vol = 382.402
core size 173 of 198  vol = 372.866
core size 172 of 198  vol = 357.001
core size 171 of 198  vol = 346.576
core size 170 of 198  vol = 337.454
core size 169 of 198  vol = 326.668
core size 168 of 198  vol = 314.959
core size 167 of 198  vol = 304.136
core size 166 of 198  vol = 294.561
core size 165 of 198  vol = 285.658
core size 164 of 198  vol = 278.893
core size 163 of 198  vol = 266.773
core size 162 of 198  vol = 259.003
core size 161 of 198  vol = 247.731
core size 160 of 198  vol = 239.849
core size 159 of 198  vol = 234.973
core size 158 of 198  vol = 230.071
core size 157 of 198  vol = 221.995
core size 156 of 198  vol = 215.629
core size 155 of 198  vol = 206.8
core size 154 of 198  vol = 196.992
core size 153 of 198  vol = 188.547
core size 152 of 198  vol = 182.27
core size 151 of 198  vol = 176.961
core size 150 of 198  vol = 170.72
core size 149 of 198  vol = 166.128
core size 148 of 198  vol = 159.805
core size 147 of 198  vol = 153.775
core size 146 of 198  vol = 149.101
core size 145 of 198  vol = 143.664
core size 144 of 198  vol = 137.145
core size 143 of 198  vol = 132.523
core size 142 of 198  vol = 127.237
core size 141 of 198  vol = 121.579
core size 140 of 198  vol = 116.78
core size 139 of 198  vol = 112.575
core size 138 of 198  vol = 108.175
core size 137 of 198  vol = 105.137
```

```
core size 136 of 198  vol = 101.254
core size 135 of 198  vol = 97.379
core size 134 of 198  vol = 92.978
core size 133 of 198  vol = 88.188
core size 132 of 198  vol = 84.032
core size 131 of 198  vol = 81.902
core size 130 of 198  vol = 78.023
core size 129 of 198  vol = 75.276
core size 128 of 198  vol = 73.057
core size 127 of 198  vol = 70.699
core size 126 of 198  vol = 68.976
core size 125 of 198  vol = 66.707
core size 124 of 198  vol = 64.376
core size 123 of 198  vol = 61.145
core size 122 of 198  vol = 59.029
core size 121 of 198  vol = 56.625
core size 120 of 198  vol = 54.369
core size 119 of 198  vol = 51.826
core size 118 of 198  vol = 49.651
core size 117 of 198  vol = 48.19
core size 116 of 198  vol = 46.644
core size 115 of 198  vol = 44.748
core size 114 of 198  vol = 43.288
core size 113 of 198  vol = 41.089
core size 112 of 198  vol = 39.143
core size 111 of 198  vol = 36.468
core size 110 of 198  vol = 34.114
core size 109 of 198  vol = 31.467
core size 108 of 198  vol = 29.445
core size 107 of 198  vol = 27.323
core size 106 of 198  vol = 25.82
core size 105 of 198  vol = 24.149
core size 104 of 198  vol = 22.647
core size 103 of 198  vol = 21.068
core size 102 of 198  vol = 19.953
core size 101 of 198  vol = 18.3
core size 100 of 198  vol = 15.723
core size 99 of 198  vol = 14.841
core size 98 of 198  vol = 11.646
core size 97 of 198  vol = 9.435
core size 96 of 198  vol = 7.354
core size 95 of 198  vol = 6.181
core size 94 of 198  vol = 5.667
```

```
 core size 93 of 198  vol = 4.706
 core size 92 of 198  vol = 3.664
 core size 91 of 198  vol = 2.77
 core size 90 of 198  vol = 2.151
 core size 89 of 198  vol = 1.715
 core size 88 of 198  vol = 1.15
 core size 87 of 198  vol = 0.874
 core size 86 of 198  vol = 0.685
 core size 85 of 198  vol = 0.528
 core size 84 of 198  vol = 0.37
 FINISHED: Min vol ( 0.5 ) reached
```

```r
core.inds <- print(core, vol=0.5)
```

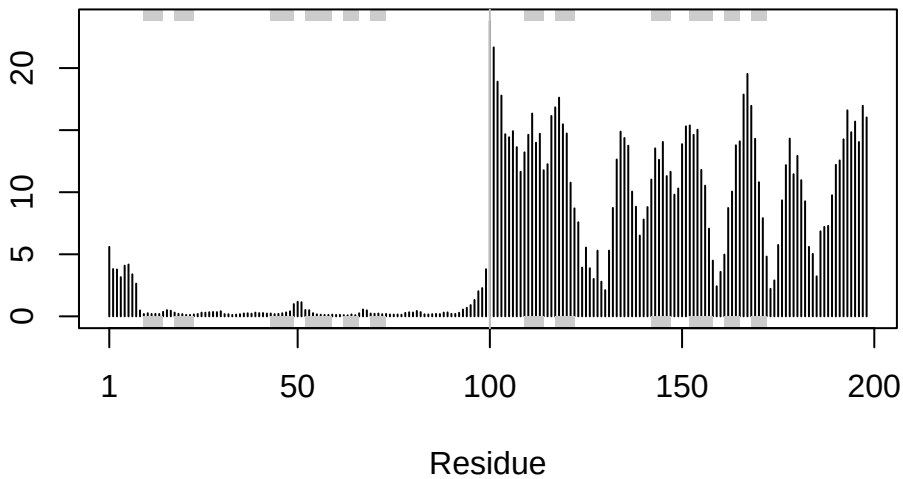
```
# 85 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     9  49     41
2    52  95     44
```

```r
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

Examine the RMSF between positions of the structure. RMSF is an often used measure of conformational variance along the structure:

```r
rf <- rmsf(xyz)

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```



Predicted Alignment Error (PAE) for Domains

Below we read these files and see that AlphaFold produces a useful inter-domain prediction for model 1 (and 2) but not for model 5 (or indeed models 3, 4, and 5):

```
library(jsonlite)

# Listing of all PAE JSON files
pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)

pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

```r
# Per-residue pLDDT scores
#   same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 90.81 93.25 93.69 92.88 95.25 89.44
```

The maximum PAE values are useful for ranking models. Here we can see that model 5 is much worse than model 1. The lower the PAE score the better. How about the other models, what are thir max PAE scores?

```r
pae1$max_pae
```

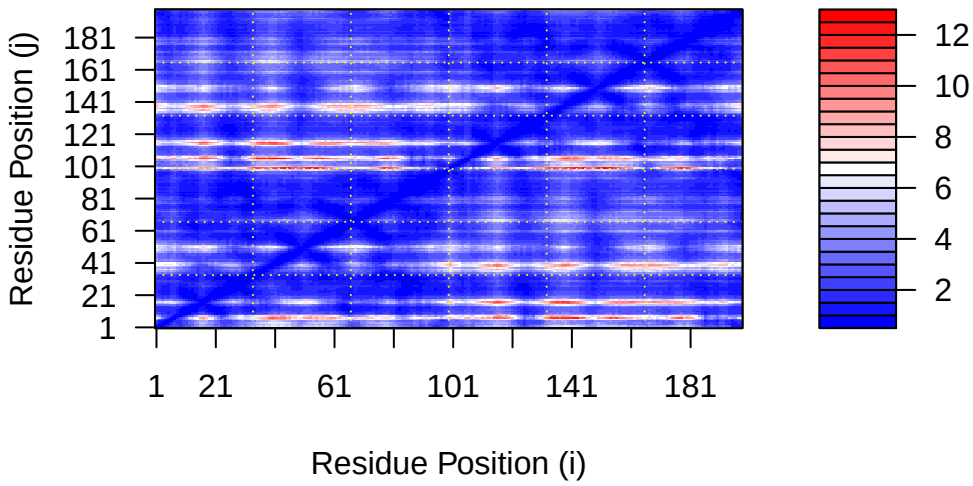
```
[1] 12.84375
```

```r
pae5$max_pae
```

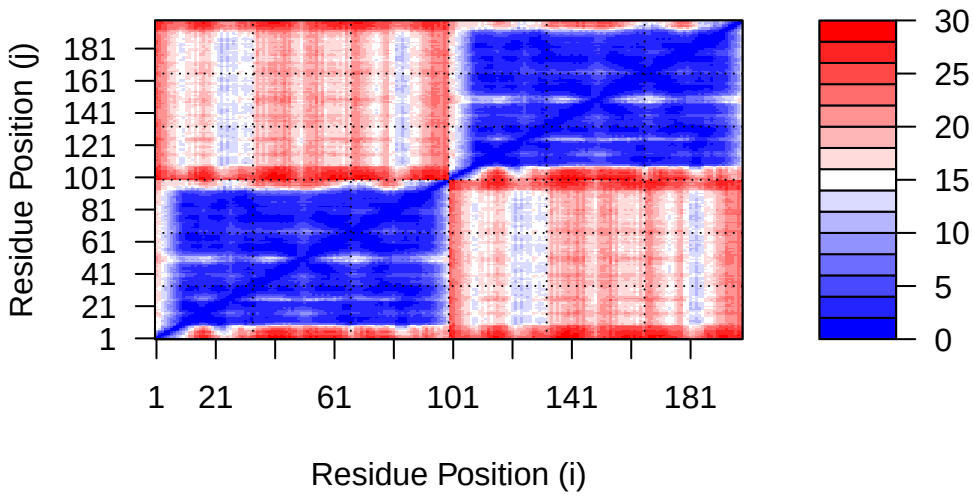
```
[1] 29.59375
```

We can plot the N by N (where N is the number of residues) PAE scores with ggplot or with functions from the `bio3d()` package:

```r
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```

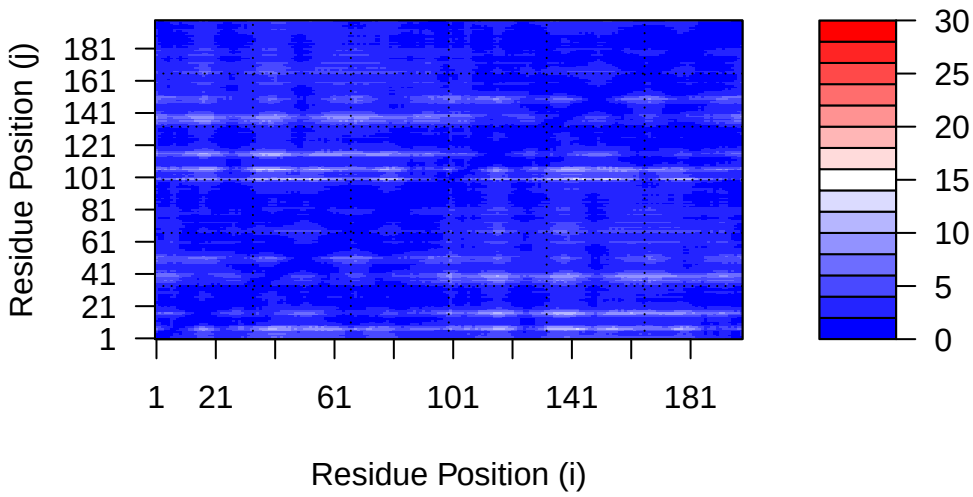



```r
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



We should really plot all of these using the same z range. Here is the model 1 plot again but this time using the same data range as the plot for model 5:

```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Residue Conservation from Alignment File

```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                        full.names = TRUE)
aln_file
```

```
[1] "hivpr_23119//hivpr_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

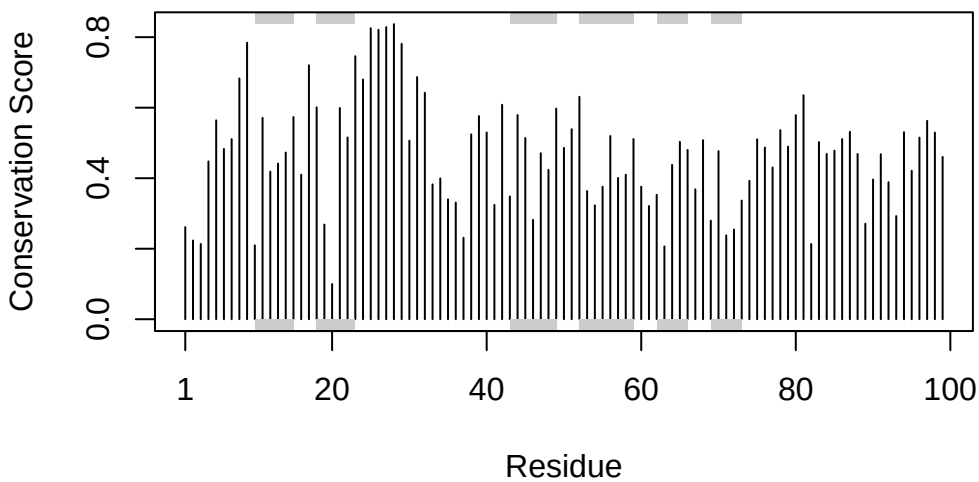
```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 5397  132
```

```r
sim <- conserv(aln)

plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```



Note the conserved Active Site residues D25, T26, G27, A28. These positions will stand out if we generate a consensus sequence with a high cutoff value:

```r
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

Make a final visualization of these functionally important sites by mapping this conservation score to the Occupancy column of a PDB file for viewing in molecular viewer programs such as Mol*, PyMol, VMD, chimera etc.

```r
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```

We can now clearly see the central conserved active site in this model where the natural peptide substrate (and small molecule inhibitors) would bind between domains.

Curent Limitations and Potential Problems

If something goes wrong with your ColabFold run, you only real option is to load the site over again. Colab notebooks can crash or time-out at any time. If you are running multiple predictions you could therefore lose a lot of work. You can mitigate this potential lose by manually downloading results as they appear (using the folder icon on the left side of the notebook, selecting a .zip file to show a download menu, and downloading the file).

Summary

In a sense AF provides all biologists with a new technique, bringing the fun of structure-gazing without the effort of experimental structure determination work. It is crucial that we educate the next generation of biologists to learn how to critically analyze predicted structures, notice new interactions, and to get to know each protein of interest in sufficient detail, so as to differentiate between "bugs" and "features".

```r
sessionInfo()
```

```
R version 4.5.2 (2025-10-31)
Platform: x86_64-apple-darwin20
Running under: macOS Sequoia 15.7.3

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/lib/libRlapack.dylib;

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Los_Angeles
tzcode source: internal

attached base packages:
```

```
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
[1] jsonlite_2.0.0  pheatmap_1.0.13 bio3d_2.4-5

loaded via a namespace (and not attached):
 [1] crayon_1.5.3        cli_3.6.5           knitr_1.51
 [4] rlang_1.1.7         xfun_0.55           otel_0.2.0
 [7] generics_0.1.4      glue_1.8.0          S4Vectors_0.48.0
[10] Biostrings_2.78.0   htmltools_0.5.9     stats4_4.5.2
[13] scales_1.4.0        rmarkdown_2.30      grid_4.5.2
[16] Seqinfo_1.0.0       evaluate_1.0.5      fastmap_1.2.0
[19] lifecycle_1.0.5     yaml_2.3.12         IRanges_2.44.0
[22] compiler_4.5.2      RColorBrewer_1.1-3  Rcpp_1.1.1
[25] XVector_0.50.0      farver_2.1.2        digest_0.6.39
[28] R6_2.6.1            parallel_4.5.2      tools_4.5.2
[31] gtable_0.3.6        msa_1.42.0          BiocGenerics_0.56.0
```